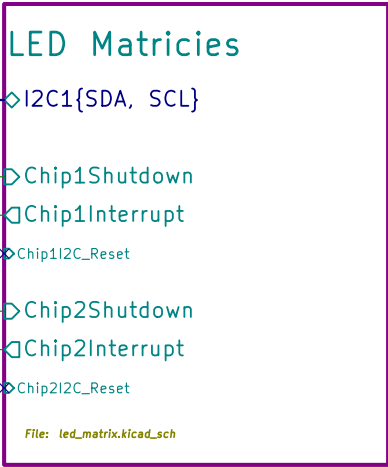
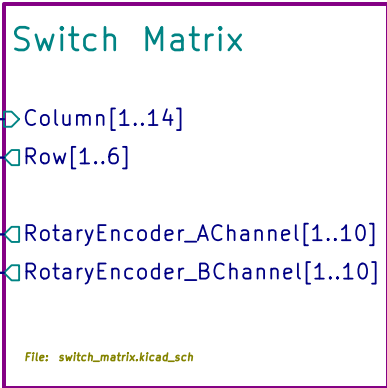
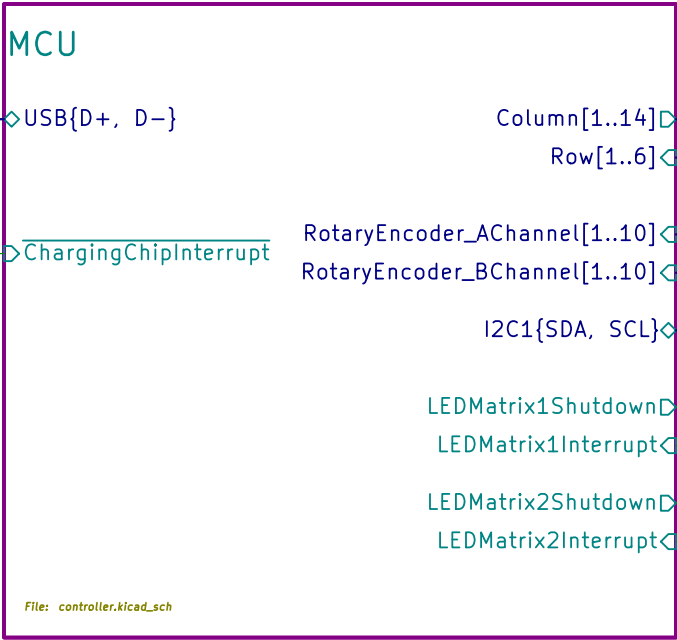
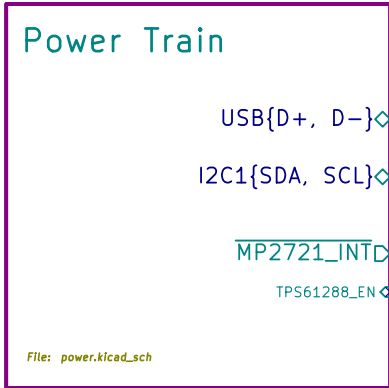




Loose Notes

- Check circuits
- I don't know if my circuit actually needs separate power and digital grounds.

- Schematic
- [] Refactor Schematic to r/PrintedCircuitBoard Guide
 - [x] Refactor the root schematic to show how the system is connected
 - I want to add logic levels and ground differentiation
 - I want custom graphics for each symbol
 - [] Refactor chip configurations to explain my choices
 - [] Rename all the component references so they are grouped and easier to layout
 - [] Select actual components, footprints, 3D models for all components
 - [] Pass those design reviews softwares

- PCB
- generate and recreate ergonomic montary key layout in PCB
 - determine latch layout on PCB

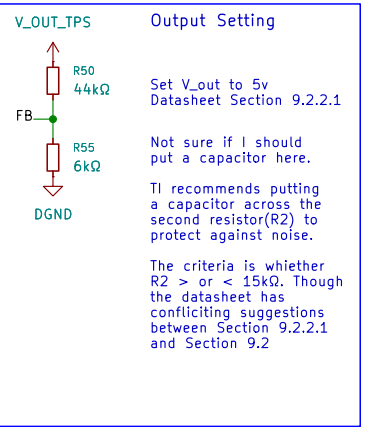
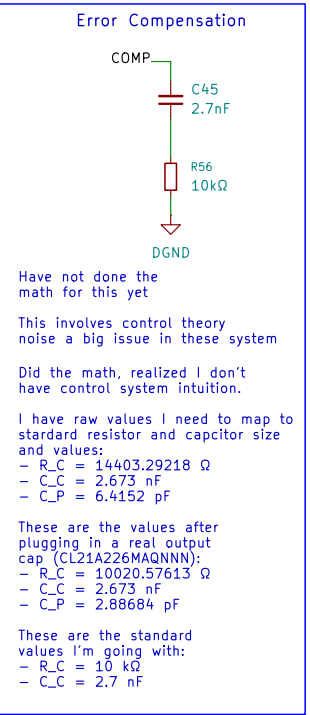
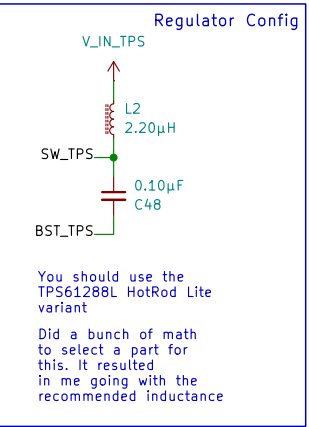
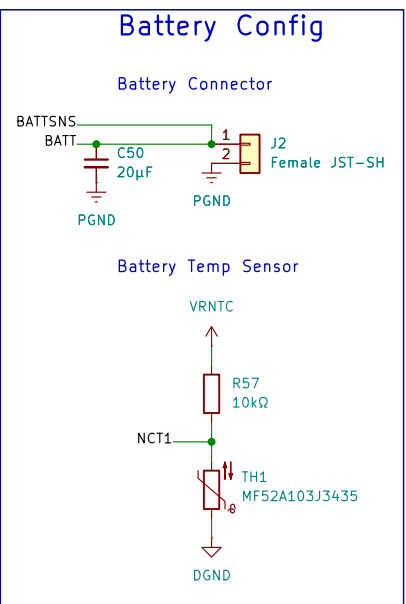
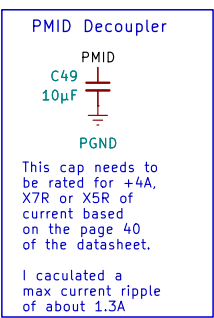
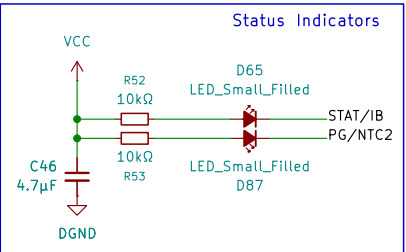
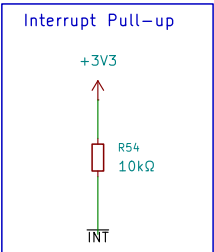
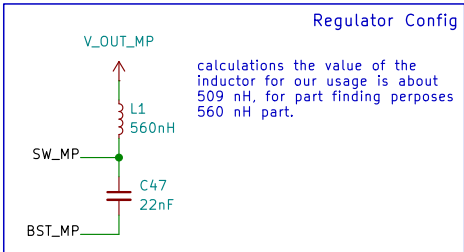
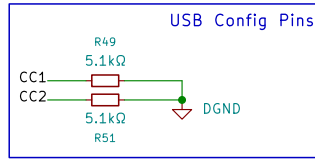
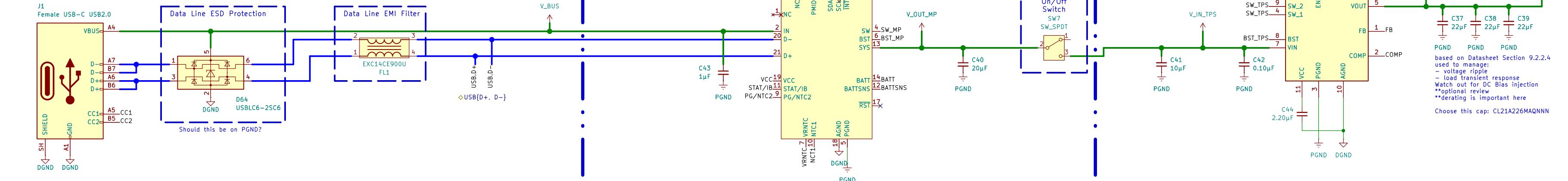
KiCad Feature Requests:
Hierarchical Bus
Hierarchical Sheet Symbols

Sheet: /		
File: slack.kicad_sch		
Title: Slack		
Size: A	Date: 2026-06-17	Rev: A.0.0
KiCad E.D.A. 10.0.1		Id: 1/7

USB-C 2.0

Charger

Boost Converter



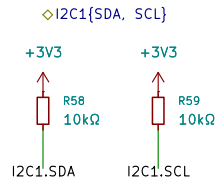
Do I have enough protect on the USB connector?
Do I still need to add resistors or pull up these lines?
Have I properly set up the charger and regulator correctly?

Is this the simplest power/usb-c module possible? Consider adding a fuel gauge ic:
Centralize the decoupling for +5V - i2c and led indicator
- BQ27220

forgot what I was putting here
Setup global reset
How to handle power and analog grounds???

Microcontroller

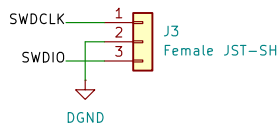
I2C Bus Config



pick these last based on the capacitance of the line on the PCB

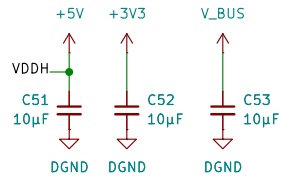
- ◇Column[1..14]
- ◇Row[1..6]
- ◇RotaryEncoder_AChannel[1..10]
- ◇RotaryEncoder_BChannel[1..10]

Programmer



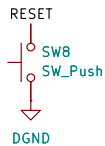
Verify that this is enough to program this board

Power Config



Regulator 0 & Regulator 1 are both set to LDO mode

Reset Button



The datasheet is unclear about behavior of this pin

Low Frequency IO Pins:
= P0.02 P0.03 P0.09 P0.10 P0.28 P0.29 P0.30 P0.31
= P1.01 P1.02 P1.03 P1.04 P1.05 P1.06 P1.07 P1.10
= P1.11 P1.12 P1.13 P1.14 P1.15

Loose Notes

Standard Frequency IO Pins:
= P0.00 P0.01 P0.04 P0.05 P0.06 P0.07 P0.08 P0.11
= P0.12 P0.13 P0.14 P0.15 P0.16 P0.17 P0.18 P0.19
= P0.20 P0.21 P0.22 P0.23 P0.24 P0.25 P0.26 P0.27
= P1.00 P1.08 P1.09

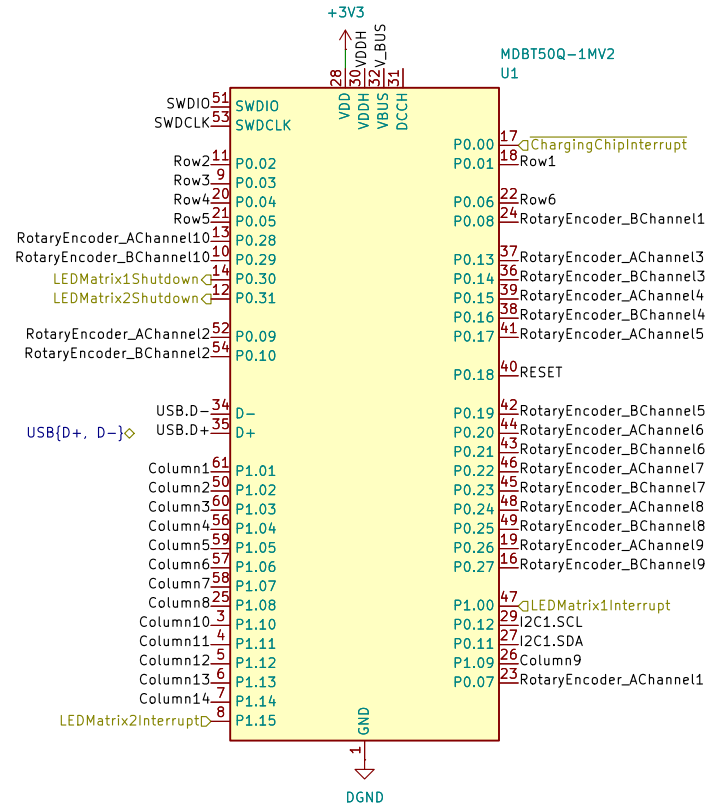
P0.18 is the nRESET Pin. The datasheet is unclear about how reset works

Be careful with the pin assignments as not every pin can handle a high speed signal

Almost any pin can be used for I2C on the nrf52. Slight this source So I'll use P0.00 and P0.01.
This might be problematic as these pins are used for the secondary clock.

Consider if there enough space for UART debug. No there will not be

Set up resets for the device.
Centralize the decoupling for VDDH



Sheet: /MCU/
File: controller.kicad_sch

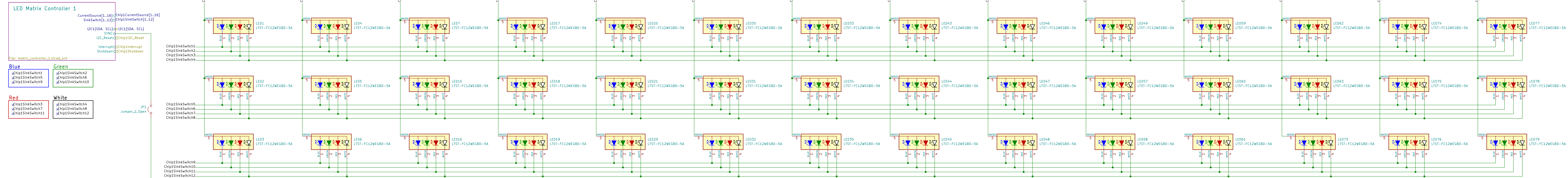
Title: Slack

Size: A Date: 2026-06-17
KiCad E.D.A. 10.0.1

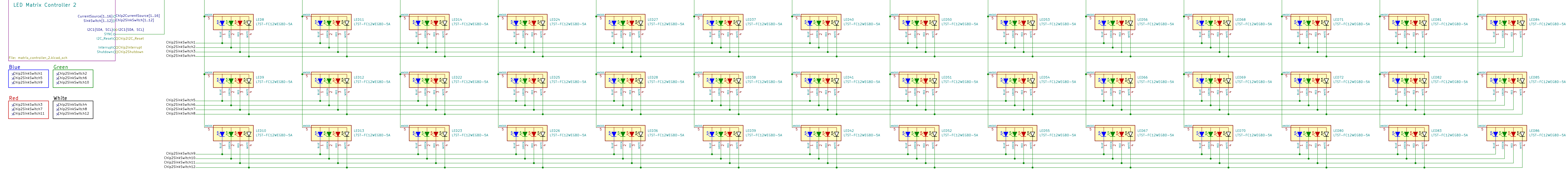
Rev: A.0.0
Id: 3/7

- Column[1..14]
- Row[1..6]
- RotaryEncoder_AChannel[1..10]
- RotaryEncoder_BChannel[1..10]

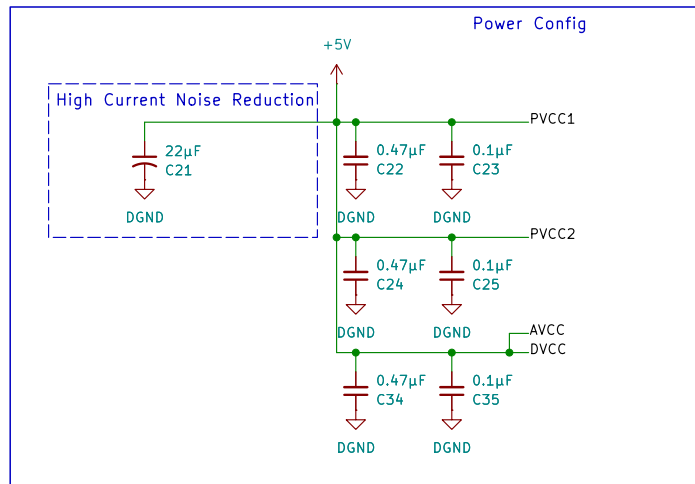
RGBW LED Matrix Halve 1



RGBW LED Matrix Halve 2



LED Matric Controller 1



◊CurrentSource[1..16]
 ◊SinkSwitch[1..12]
 ◊I2C1{SDA, SCL}
 ◊SYNC
 ◊I2C_Reset
 ◊Shutdown
 ◊Interrupt

Blue

✕SinkSwitch1
 ✕SinkSwitch5
 ✕SinkSwitch9

Green

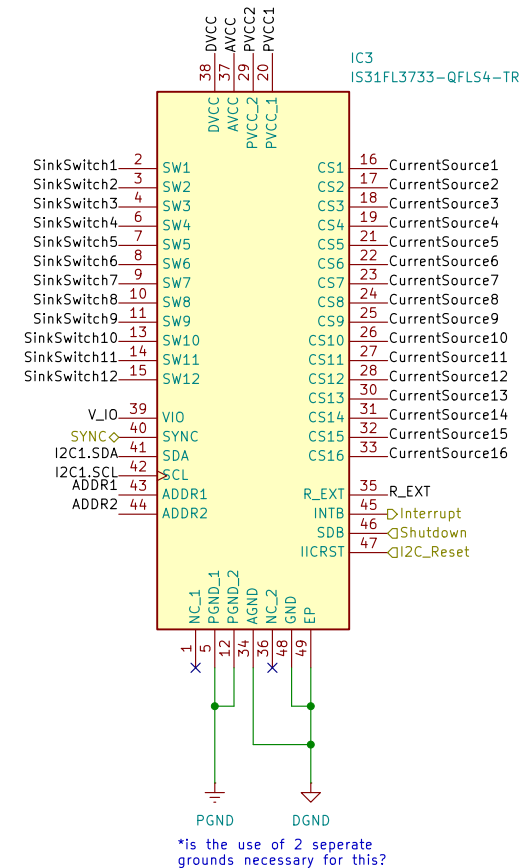
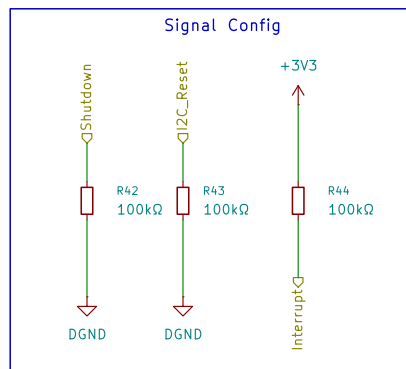
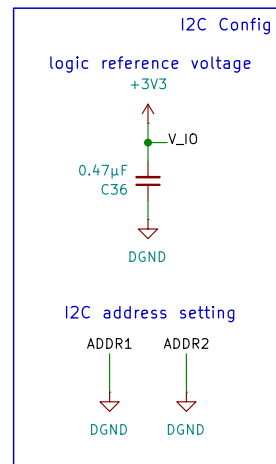
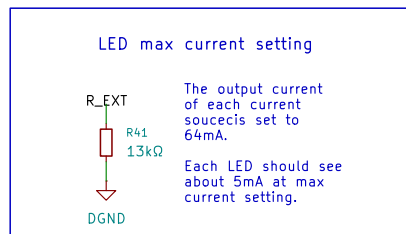
✕SinkSwitch2
 ✕SinkSwitch6
 ✕SinkSwitch10

Red

✕SinkSwitch3
 ✕SinkSwitch7
 ✕SinkSwitch11

White

✕SinkSwitch4
 ✕SinkSwitch8
 ✕SinkSwitch12



Sheet: /LED Matrices/LED Matrix Controller 1/
File: matric_controller_1.kicad_sch

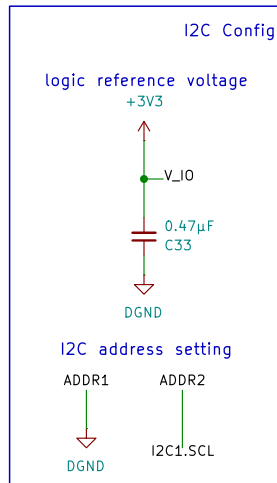
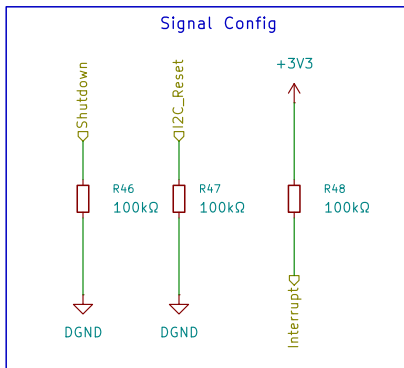
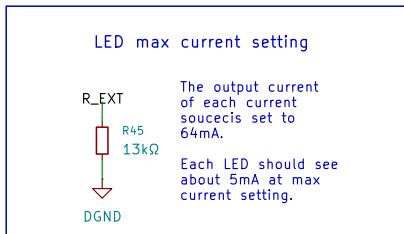
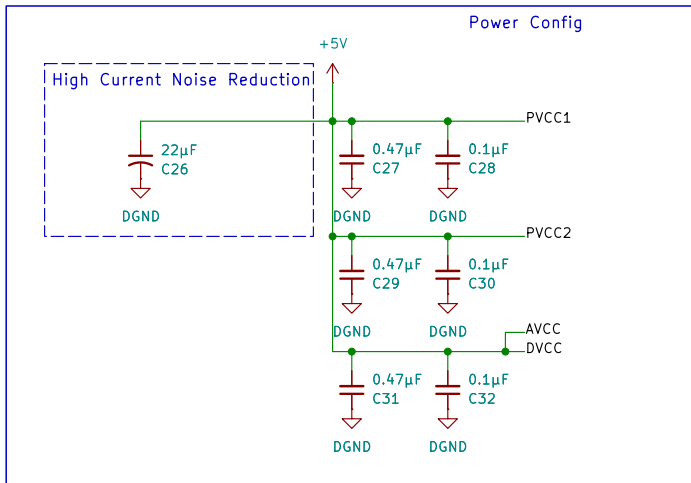
Title:

Size: A Date:

KiCad E.D.A. 10.0.1

Rev:
Id: 6/7

LED Matric Controller 1



◊CurrentSource[1..16]
 ◊SinkSwitch[1..12]
 ◊I2C1{SDA, SCL}
 ◊SYNC
 ◊I2C_Reset
 ◊Shutdown
 ◊Interrupt

Blue

>SinkSwitch1
 >SinkSwitch5
 >SinkSwitch9

Red

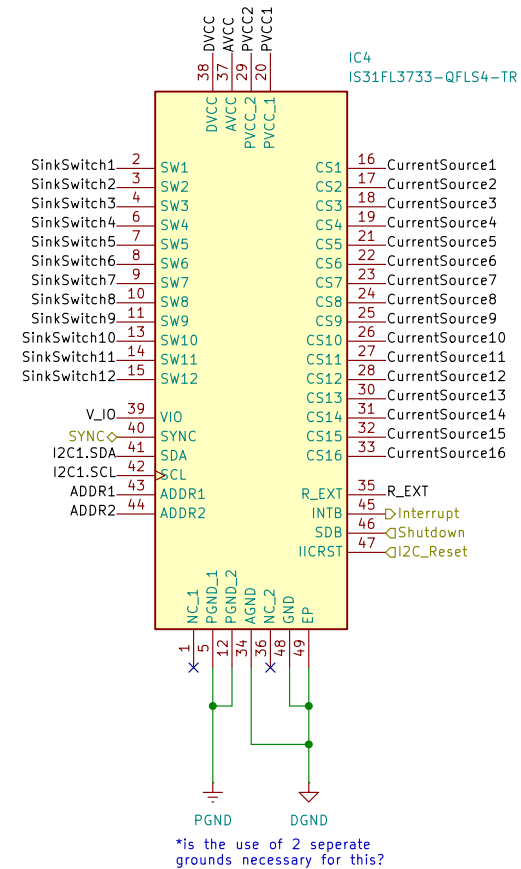
>SinkSwitch3
 >SinkSwitch7
 >SinkSwitch11

Green

>SinkSwitch2
 >SinkSwitch6
 >SinkSwitch10

White

>SinkSwitch4
 >SinkSwitch8
 >SinkSwitch12



Sheet: /LED Matrices/LED Matrix Controller 2/
File: matrix_controller_2.kicad_sch

Title:

Size: A Date:

KiCad E.D.A. 10.0.1

Rev:

Id: 7/7